Course Project Progress Report

CSE 578: Data Visualization

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Problem Statement

UVW College is looking to bolster its enrollment by targeting specific demographics based on income level. To support this effort, UVW College's marketing team has enlisted XYZ Corporation to assist them with this project. UVW aims to focus on individuals earning less than $50,000 annually as the primary criterion for marketing its degree programs. Our task at XYZ is to work with UVW to use U.S. Census Bureau data to determine the key demographics to target in the marketing campaign. The dataset includes key variables that help assess whether individuals earn less than or more than $50,000, such as age, work class, education level, marital status, and occupation.

To achieve its enrollment target, the marketing team at UVW would like XYZ to research and identify important factors influencing salary. With this information, the UVW marketing team can use the information to efficiently and effectively target their marketing efforts to the demographic groups that can most benefit from having a college degree. The groups with the most benefits might have a higher incentive to attend UVW College.

Progress Made

The project requirements and dataset from the UCI Machine Learning Repository were reviewed to understand available data and its representation. The dataset was examined to determine the available columns and to assess their relationship with income levels. Several fields, including education, education years, age, and marital status, were transformed into broader grouping categories to improve summarization and analysis. These transformations enabled a more effective examination of demographic factors influencing income by reducing complexity while preserving important information.

Different ways to display the data were investigated to better understand the strengths and weaknesses of each type. Based on different types of information to be communicated, bar charts, pie charts, stack bar charts, column charts, and line charts were explored. Understanding how to effectively use charts to visually explain how factors like education, age, and marital status affect income is vital to the project.

Issues Encountered

One of the main challenges encountered during data exploration was determining which factors were most influential in differentiating individuals earning above and below $50,000. Initially, individual demographic factors such as education, years of education, age, marital status, and work class were examined to understand their impact on salary. Once multiple factors were grouped, the number of combinations got out of hand. To better handle the growth of factor combinations, several data transformations were introduced to improve grouping and segmentation. Age was categorized into three intervals: under 25, 26-30, and 31 and above. Education was consolidated into three broad categories: high school or less, some college, and a bachelor's degree or higher.

Marital status was grouped into married and not married to streamline comparisons. Transformations help to better understand the relationship between factors and income by preserving important factor information without overwhelming the analysis with an overly large number of values.

Completed Tasks

The dataset was analyzed to identify key demographic factors influencing income levels. This analysis involved exploring the correlation between variables such as age, education, marital status, and salary. Several data transformations were applied to consolidate information in different categories as described in the Issues Encountered section of the paper to determine its relation to income.

Relevant factors influencing income, including education, work class, and marital status, were identified. These insights will guide UVW College's marketing strategy by highlighting which groups will most likely benefit from further education. Initial visualizations, such as bar and stack charts, were created to compare income across different demographic groups. The work on user stories and associated visualizations is still in progress.

Tasks To Complete

Finish analyzing the data and create the needed stories and visualizations. The final report will include a definition of the goals and business objectives. The report will also contain at least five user stories prioritized for the project. Across the five user stories, a minimum of 8 out of the 14 attributes must be used. Of the five visualizations, at least three must be multivariate, while no more than two may be univariate. Each visualization must be thoroughly explained, detailing what it demonstrates, why it was chosen, the design process, and conclusions drawn from it.

Additional sections in the report will document questions that arose during the project progression, along with the solutions implemented to address them. Any features or analyses that are not currently included but are planned will be listed to help guide future development priorities. Finally, the report will be reviewed to ensure all written content, visualizations, and analyses effectively address UVW College's objectives and align with project requirements.